

Agentic AI

What is an LLM?

An LLM (Large Language Model) is like a very knowledgeable person who can talk and write but cannot take actions in the real world. When you ask ChatGPT something, it generates text based on its training. It can tell you how to book a hotel, write code for a login page, or explain concepts. But it cannot actually book the hotel, run the code, or interact with external systems.

LLM = Information Provider

LLM: chatgpt, text generate, no do any work just generate content.

LLM give information (code,hotels in lahore)

What is AI agents:

Definition

Agentic AI refers to AI systems that can act autonomously to achieve goals. Unlike traditional AI that just responds to queries, agentic AI can:

- Plan multi-step tasks
- Make decisions
- Take actions
- Learn from outcomes
- Work independently with minimal human intervention

Traditional AI waits for instructions. Agentic AI figures out what needs to be done. When you give an agent a goal like "Plan my vacation to Japan," it does not just list tourist spots. It searches for flights, checks hotel availability, compares prices, books everything, adds events to your calendar, and sends you the complete itinerary. This represents a shift from AI that talks to AI that does. Instead of humans copying code from ChatGPT and running it themselves, agents can write, test, and deploy code independently. Like an AI that just give code vs an Agent who write, test and deploy code give complete app.

Agent: generate content + work

Agent=information+Agents

Agent = LLM + Tools + Memory + Planning + Action

Characteristics of AI Agents

1.Autonomy?:

Autonomy means the agent can make decisions and take actions without constant human guidance. An autonomous agent receives a goal and figures out the steps needed to achieve it. It does not ask for permission at every step.

There are different levels of autonomy. Human-in-loop means the agent asks before taking important actions. Human-on-loop means the agent works independently but humans can intervene if needed. Fully autonomous means the agent completes tasks without any human involvement.

What they need to do decide themselves (need human in loop, again ask to LLM, do work)

2.Planning iteration and feedback:

Agents do not jump straight to action. They first create a plan by breaking the goal into smaller steps. Then they execute each step and check the results. If something goes wrong, they revise the plan and try again.

This cycle continues until the goal is achieved. For example, if an agent is writing code and gets an error, it reads the error message, understands what went wrong, fixes the code, and runs it again. This iterative process continues until the code works.

3.Coordination(sometime):

Agent1(plan) —give control to next agent—— Agent2——Agent3

Complex tasks often require multiple agents working together. One agent might plan the overall approach. Another agent might execute specific steps. A third agent might verify the results. These agents communicate and coordinate to complete the task.

Think of it like a team where each member has a specific role. The planner decides what needs to be done. The worker executes the tasks. The reviewer checks quality. Together they achieve more than any single agent could alone.

How Agent work and component:

Agent

Natural Language processing: ik fun page bana do , i want fun page, create a fun page

1.Large Language Model (The Brain)

The LLM is the thinking component of the agent. It understands natural language, reasons about problems, and makes decisions. When a user says "create a login page," the LLM understands what a login page is and what steps are needed to create one.

Popular LLMs used in agents include GPT-4, Claude, Gemini, and Llama. The LLM provides the reasoning capability that makes agents intelligent.

2.Memory:

Memory allows agents to remember information across interactions. Without memory, every conversation starts fresh. With memory, agents remember user preferences, past conversations, and previous actions.

Short-term memory holds the current conversation context. Long-term memory stores user preferences and important information permanently. Episodic memory remembers past experiences and tasks. This allows agents to learn and improve over time

3.Tools (The Capabilities)

Tools are external functions that agents can use to take actions. An LLM alone can only generate text. With tools, it can search the internet, run code, access databases, send emails, and interact with APIs.

Each tool is like a function in programming. The agent decides which tool to use based on the task. For example, to check weather, it uses a weather API. To send a message, it uses an email or messaging tool. To analyze data, it uses a code interpreter.

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tool(toolbox,code ma functions for example create_a_logon_page(data){})

4.Planning (The Decision Maker)

Planning is how agents decide what steps to take. The agent receives a goal and breaks it into smaller, manageable tasks. It decides the order of execution and which tools to use for each step.

Good planning prevents wasted effort. An agent that plans well will not search for hotels before knowing the travel dates. It sequences actions logically to achieve the goal efficiently.

5.Action (The Execution)

Action is where the agent actually does something. After planning and tool selection, the agent executes the actions. This could mean making API calls, writing files, sending messages, or any other concrete action.

The action component is what separates agents from regular LLMs. An LLM can describe how to send an email. An agent actually sends the email.

Agentic workflow:

User query -----<Make a plan>----- Planning agent ----- > EXecute action with tool -<result no ok>-----

reflect onResult----- result okay—response

User input, Agent1: Agent 2 Agent3

 Classifier Agent Selection tool response

User get response conversational history (memory)

Agentic AI v generative AI

Generative AI, as its name suggests, is focused on the creation of new content, such as text, images, code, or music, based on input prompts. The LLM is at the heart of generative AI, and the value is generated by what the model can do and simple extensions of the LLM's capabilities. For example, you can generate or edit content, and even perform simple function calling and chain together various options.

Agentic AI is a subset of generative AI that is centered around the orchestration and execution of agents that use LLMs as a "brain" to perform actions through tools. Agentic AI goes beyond content creation and function calling by executing actions in underlying systems to achieve higher-level goals.

For example, generative AI could be used to create marketing materials, while agentic AI could then be used to deploy these materials, track their performance, and automatically adjust the marketing strategy based on the results. In this way, agentic AI can use generative AI as a tool to achieve its goals.

Generative AI:content creation

Agent AI: decision making and execution(tools)

Types of AI Agents

Simple Reflex Agents

These are the most basic agents. They react to current input without any memory or planning. When they receive input, they apply predefined rules and produce output.

A spam filter is a simple reflex agent. It looks at each email and decides if it is spam based on rules. It does not remember past emails or plan anything.

Model-Based Reflex Agents

These agents maintain an internal model of the world. They remember past perceptions and use that information to make decisions. This allows them to handle situations where current input is incomplete.

A chatbot with conversation history is model-based. It remembers what was said earlier and uses that context to understand current messages.

Goal-Based Agents

These agents work towards specific objectives. They do not just react to input. They plan actions that move them closer to their goal.

A navigation app is goal-based. The goal is to reach a destination. The agent plans a route and provides directions to achieve that goal.

Single vs Multi-Agent Systems

Single Agent Systems

A single agent handles all aspects of a task alone. It plans, executes, and verifies everything itself. This is simpler but may be limited for complex tasks.

A personal assistant chatbot is typically a single agent. It handles all user requests independently.

Multi-Agent Systems

Multiple agents work together to complete tasks. Each agent may have different capabilities or specializations. They communicate and coordinate to achieve common goals.

This is like a team where each member has expertise in different areas. Together they can handle more complex tasks than any single agent.

Hierarchical Systems

In hierarchical systems, there is a manager agent that coordinates worker agents. The manager receives the goal, delegates tasks to workers, and combines their results.

This is useful for large projects where coordination is important. The manager ensures all pieces fit together properly.

Memory Systems in Agents

Short-Term Memory

Short-term memory holds information for the current session. It includes the ongoing conversation and immediate context. When the session ends, this memory is cleared.

This is like your working memory when solving a problem. You keep relevant information in mind while working.

Long-Term Memory

Long-term memory persists across sessions. It stores user preferences, important facts, and learned information. This allows agents to remember users and personalize interactions.

This is like your long-term memory where you store important life information.

Episodic Memory

Episodic memory stores specific experiences and events. It remembers what happened in past tasks and can recall those experiences when relevant.

This helps agents learn from specific situations they have encountered before.

Semantic Memory

Semantic memory stores general knowledge and facts. This includes concepts, relationships, and understanding of how things work.

This is the agent's knowledge base that it uses to understand the world.

Procedural Memory

Procedural memory stores skills and how-to information. It remembers how to perform specific tasks and workflows.

This allows agents to execute familiar tasks efficiently without relearning each time.